Visual Question Answering using Pretrained Vision-Language Models

Course: Language Processing using ML and LLMs **Task:** 14 (VQA)

1. Title and Group Details

Name	Roll No	Contribution
Aditya Srivastava	2330062	Individual project: design, implementation, evaluation, report

2. Problem Statement and Objectives

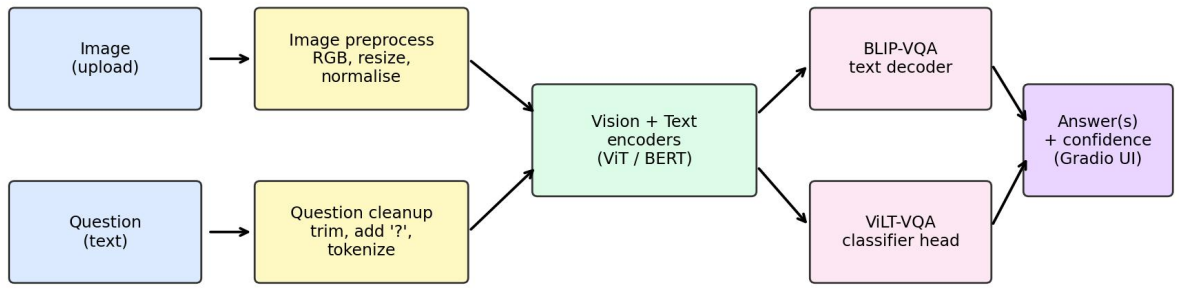
Visual Question Answering (VQA) takes an image and a free-form question and returns a short answer. It joins computer vision and language understanding and supports accessibility tools for visually impaired users, image retrieval, education and content moderation.

Objectives:

- Build a working VQA application using pretrained models.
- Compare a generative model (BLIP) with a classification-based model (ViLT).
- Evaluate accuracy by question type and analyse failures.
- Provide a simple web interface and a reproducible repository.

Expected outcome: a Gradio app that answers questions about any uploaded image, plus an evaluation table.

3. Methodology and Processing Pipeline



Input: an image (JPG/PNG) and a text question.

Preprocessing: the image is converted to RGB, resized and normalised by the model processor; the question is trimmed, a question mark is added if missing, and it is tokenised.

Intermediate processing: a vision transformer encodes image patches and a text encoder encodes the question; cross-modal attention fuses them.

Output: BLIP generates an answer with a text decoder; ViLT picks the best of 3,129 candidate answers and gives a sigmoid confidence. The interface shows both answers side by side.

4. Models and Tools Used

Model	Source	Type
BLIP-VQA base	huggingface.co/Salesforce/blip-vqa-base	Generative encoder-decoder
ViLT-B/32 VQA	huggingface.co/dandelin/vilt-b32-finetuned-vqa	Classifier over 3,129 answers

Tools: Python 3.10+, PyTorch, Hugging Face Transformers, Pillow, Pandas, Gradio. Environment: Google Colab or local machine (CPU or GPU).

5. Implementation and Usability

- **vqa_pipeline.py:** wraps both models behind one answer(image, question) method with cached loading.
- **app.py:** Gradio interface with image upload, question box and model selection.
- **evaluate.py:** runs both models on samples/questions.csv, computes normalised exact-match accuracy per model and per question type (yes/no, number, other), and records latency.
- **To run:** pip install -r requirements.txt; python src/app.py; python src/evaluate.py samples/questions.csv.
- **Applications:** assistive description tools, quick image inspection, teaching aid, retail and e-commerce image queries.

6. Results and Sample Outputs

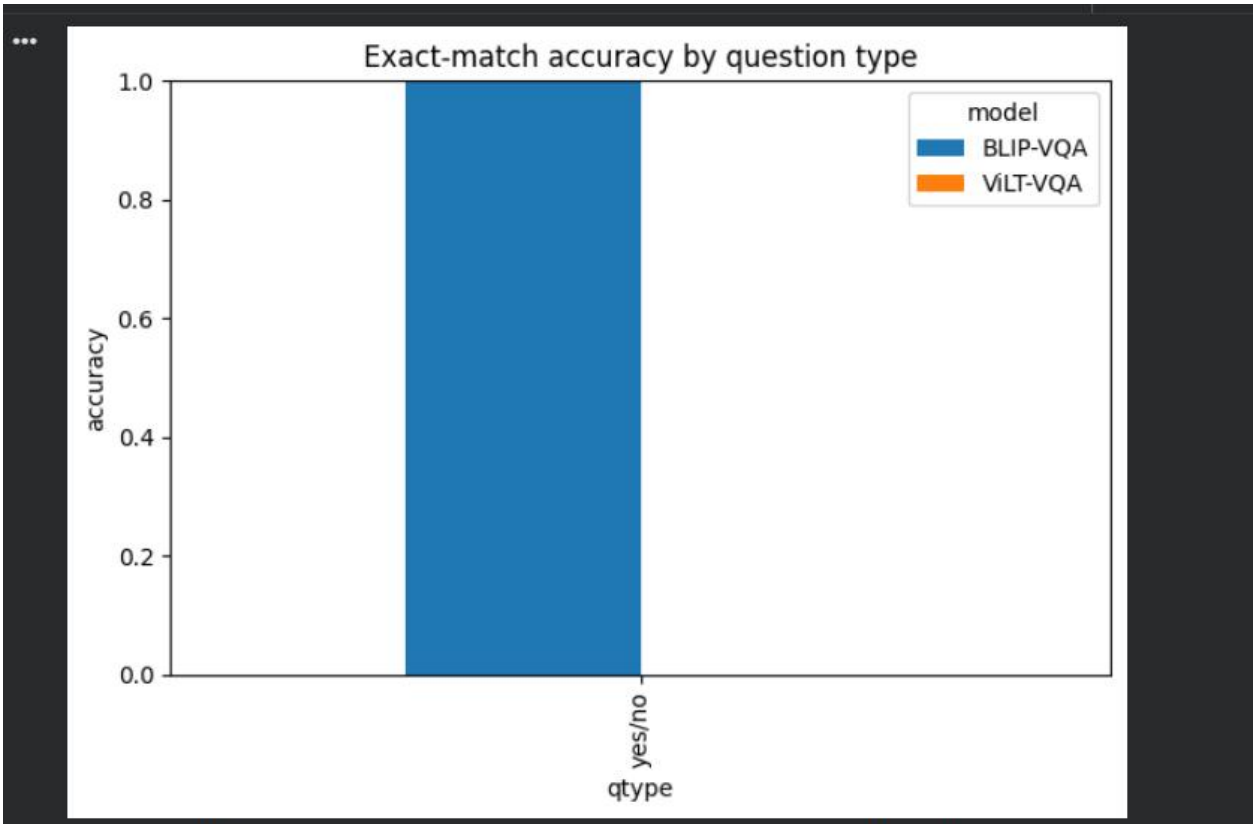
```
         correct    sec
model
BLIP-VQA     1.0  22.397
ViLT-VQA     0.0   3.835
model   BLIP-VQA  ViLT-VQA
qtype
yes/no      1.0      0.0
```

	model	image	question	gold	pred	correct	qtype	sec
0	BLIP-VQA	jason-strull-FSiQFlwnwfw-unsplash.jpg	Is it night time?	yes	yes	True	yes/no	22.397308
1	ViLT-VQA	jason-strull-FSiQFlwnwfw-unsplash.jpg	Is it night time?	yes	no	False	yes/no	3.835030

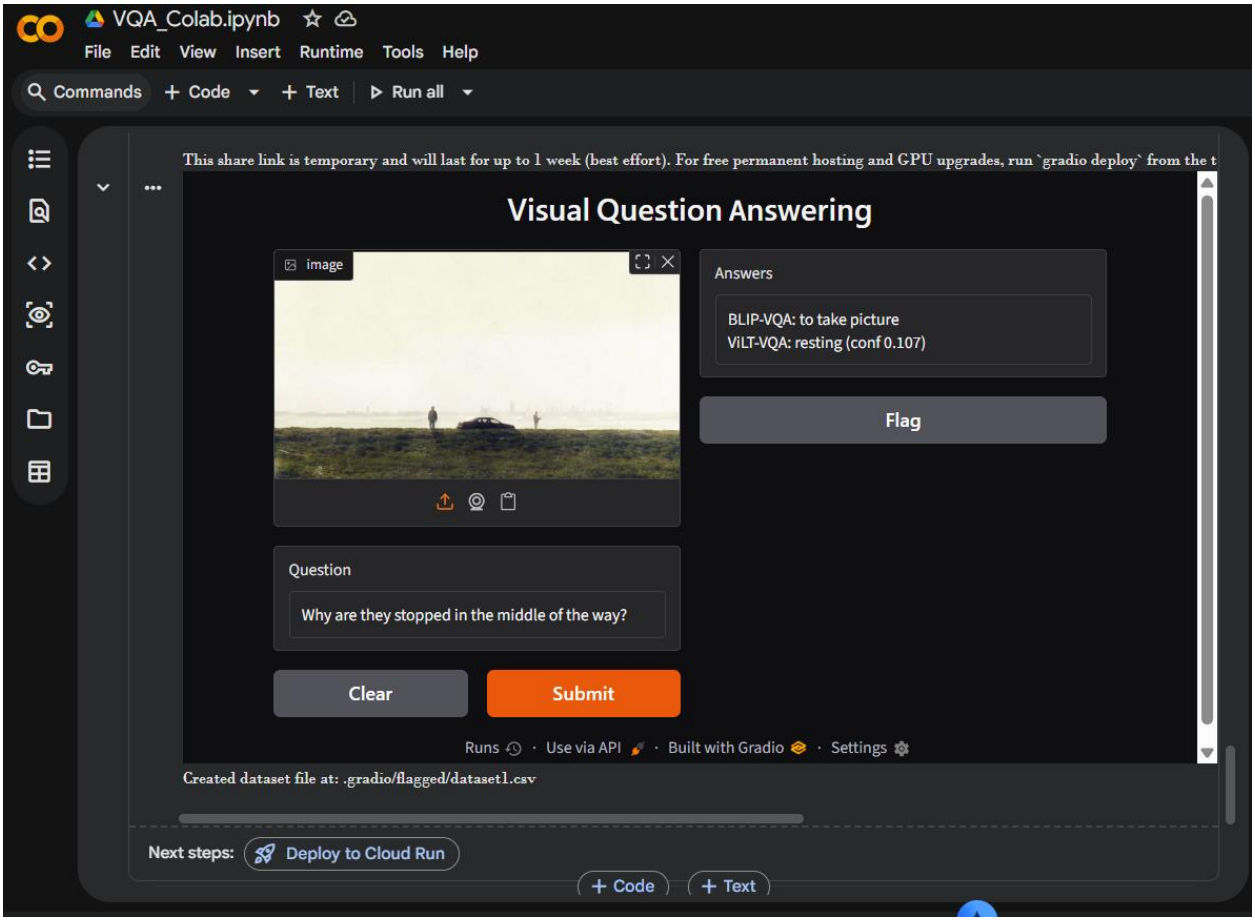
Observations to discuss (verify against your results):

- Compare how each model handles yes/no versus open-ended questions.
- Check whether counting questions beyond small numbers fail.
- Note differences in answer style: BLIP may produce free text, ViLT only answers from its fixed vocabulary.
- Report which model is faster on your hardware.

Limitations: trained mainly on everyday photographs (VQA v2 style); weak at counting, reading text in images, spatial reasoning and Indian cultural content; ViLT cannot output answers outside its 3,129-answer vocabulary; answers carry no explanation; accuracy is measured on a small custom test set, so results are indicative only.



	model	image	question	gold	pred	correct	qtype	sec
1	ViLT-VQA	jason-strull-FSiQFlwnwfw-unsplash.jpg	Is it night time?	yes	no	False	yes/no	3.83503



7. Conclusion and Future Scope

The project delivers a working VQA application built on two pretrained models, with a reproducible evaluation. Future work: fine-tune on domain data (for example Indian heritage images), add larger multimodal LLMs such as LLaVA, support Hindi questions via translation, add answer explanations and highlight image regions.

8. References and GitHub Repository

- Li et al., BLIP: Bootstrapping Language-Image Pre-training, ICML 2022.
- Kim et al., ViLT: Vision-and-Language Transformer Without Convolution or Region Supervision, ICML 2021.
- Goyal et al., Making the V in VQA Matter (VQA v2), CVPR 2017.
- Hugging Face Transformers documentation: https://huggingface.co/docs/transformers
- **GitHub repository:** https://github.com/Adityashri277/VQA